Assignment 3 — SPICE & LTspice

Revision questions, accepted answers, and concise explanations

Q1. The acronym SPICE stands for:

A. System Program for Integrated Circuit Electronics
B. Simulation Process for Industrial Circuit Engineering
C. Simulation Program with Integrated Circuit Emphasis
D. Standard Program for Intelligent Circuit Evaluation

Correct answer: C. Simulation Program with Integrated Circuit Emphasis

Explanation: SPICE is a general-purpose circuit simulation program originally developed to analyze electronic circuits using mathematical models.

Exam tip: SPICE = Simulation Program with Integrated Circuit Emphasis.

Q2. LTspice was originally developed by:

A. Intel
B. Texas Instruments
C. Cadence Design Systems
D. Linear Technology

Correct answer: D. Linear Technology

Explanation: LTspice was developed by Linear Technology as a SPICE-based simulation tool. Linear Technology was later acquired by Analog Devices.

Exam tip: For this quiz, the accepted answer is Linear Technology.

Q3. Which analysis is used to determine the steady-state voltages and currents in a circuit?

A. AC Analysis
B. Operating Point Analysis
C. Transient Analysis
D. Noise Analysis

Correct answer: B. Operating Point Analysis

Explanation: Operating-point analysis calculates the circuit's DC bias conditions: node voltages and branch currents. It is commonly used to establish the initial bias point for other analyses.

Exam tip: Operating point = DC bias voltages and currents.

Q4. One important advantage of simulation before hardware implementation is that it:

A. Eliminates the need for semiconductor fabrication
B. Reduces the need to verify circuit behavior
C. Helps verify circuit operation before physical implementation
D. Removes the need for component models

Correct answer: C. Helps verify circuit operation before physical implementation

Explanation: Simulation lets a designer check expected behavior, explore operating conditions, and identify potential issues before building physical hardware. It supports testing but does not eliminate the need for hardware validation.

Exam tip: Simulation reduces design risk and can save prototyping time and cost.

Q5. A designer wants a simulation to remain unchanged even if the device manufacturer updates its online SPICE model in the future. Which import method best satisfies this requirement?

A. Linking directly to the manufacturer's URL
B. Using the built-in LTspice diode library only
C. Including a locally stored model file in the project
D. Copying only the model name into the schematic

Correct answer: C. Including a locally stored model file in the project

Explanation: A local copy keeps the simulation tied to a specific version of the model rather than depending on a remote file that may change or become unavailable.

Exam tip: For reproducibility, keep the model file with the project and record its version/source.

Q6. Which sequence best reflects the workflow for using a third-party SPICE model in LTspice?

A. Create schematic → Obtain manufacturer model → Check existing library → Assign downloaded model → Simulate
B. Create schematic → Check existing library → Obtain manufacturer model → Assign downloaded model → Simulate
C. Check existing library → Create schematic → Assign downloaded model → Obtain manufacturer model → Simulate
D. Create schematic → Assign downloaded model → Check existing library → Obtain manufacturer model → Simulate

Correct answer: B. Create schematic → Check existing library → Obtain manufacturer model → Assign downloaded model → Simulate

Explanation: First build the schematic and check whether a suitable model is already available. If not, obtain the manufacturer's model, link or assign it correctly, then run and inspect the simulation.

Exam tip: Check the existing library before downloading an external model.

Q7. Why is knowledge of capacitor charge and inductor current essential before transient analysis begins?

A. They determine the circuit's initial conditions
B. They define the DC Sweep increment
C. They specify the semiconductor model
D. They determine the PCB layout constraints

Correct answer: A. They determine the circuit's initial conditions

Explanation: Capacitor voltage is related to stored charge, and inductor current represents stored magnetic energy. Their initial values influence the circuit's response immediately after a switching event.

Exam tip: Capacitor voltage and inductor current cannot change instantaneously in ideal components.

Q8. When evaluating a diode for a specific application, which pair of specifications must both satisfy the circuit requirements?

A. Package type and manufacturer
B. Reverse-voltage rating and forward-current rating
C. Model name and webpage URL
D. Junction temperature and simulation step size

Correct answer: B. Reverse-voltage rating and forward-current rating

Explanation: The diode must withstand the maximum reverse voltage and safely carry the required forward current. Other specifications, such as power dissipation and recovery time, may also matter depending on the application.

Exam tip: Check reverse-voltage and forward-current limits first; then verify power and thermal limits.

Q9. Which sequence correctly represents the thermal relationship discussed for estimating junction temperature?

A. Ambient temperature → Power dissipation → Thermal resistance → Junction temperature
B. Junction temperature → Ambient temperature → Thermal resistance → Power dissipation
C. Thermal resistance → Junction temperature → Ambient temperature → Power dissipation
D. Power dissipation → Junction temperature → Ambient temperature → Thermal resistance

Correct answer: A. Ambient temperature → Power dissipation → Thermal resistance → Junction temperature

Explanation: A common steady-state estimate is $T_J = T_A + P_D \times \theta_{JA}$, where T_J is junction temperature, T_A is ambient temperature, P_D is dissipated power, and θ_{JA} is junction-to-ambient thermal resistance.

Exam tip: Formula: $T_J = T_A + P_D \theta_{JA}$. Keep temperature, power, and thermal resistance units consistent.

Q10. Which combination of information is essential before deciding whether a component can be reliably used in a circuit?

A. Manufacturer name and package style
B. Electrical ratings and thermal characteristics
C. SPICE model name and simulation software version
D. Package dimensions and PCB footprint

Correct answer: B. Electrical ratings and thermal characteristics

Explanation: A component must operate within its voltage, current, power, and temperature limits. Thermal behavior is important because excessive junction temperature can reduce reliability or damage the component.

Exam tip: Always verify electrical limits and thermal limits against the circuit's worst-case operating conditions.
